

**REFINED QUALITY ASSURANCE TESTING
DOCUMENTATION (QATD)**

UMHackathon 2026

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Document Control

Field	Detail
System Under Test (SUT)	PeopleGraph – AI Decision Support System for SMEs
Team Repo URL	https://github.com/TeohXiXian-2025/umhackathon-1
Project Board URL	https://github.com/users/TeohXiXian-2025/projects/1/views/1
Live Deployment URL	https://umhackathon-1-9n2r.vercel.app/

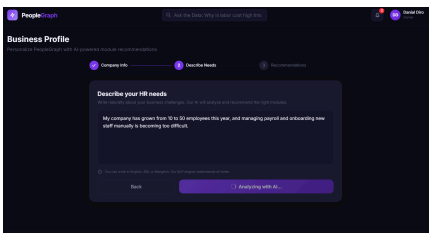
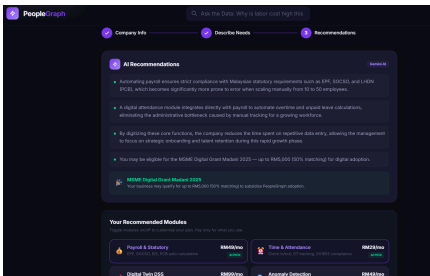
Objective:

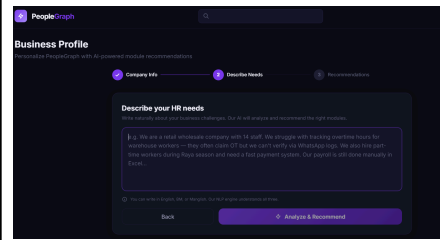
The objective of this refined QATD is to validate that **PeopleGraph functions as a true Decision Intelligence System (not just an HR tool)** by ensuring:

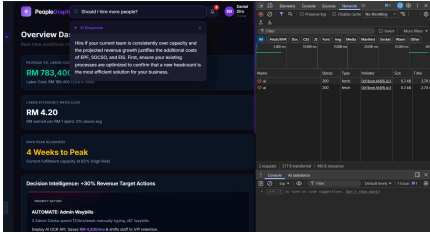
- Accurate Malaysian statutory compliance (EPF, SOCSO, EIS, PCB)
- Reliable AI-driven workforce decision support using ILMU-GLM-5.1
- Stability of Digital Twin simulation under realistic SME workloads
- Proper handling of structured + unstructured data (Payroll + WhatsApp/Manglish)
- System resilience under failure conditions (API, model, or data loss)

FINAL ROUND (Execution, RCA & Hard Evidence)

1. Test Execution & Evidence (Manual)

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
TC-01	Happy Case (Entire Flow): SME HR Needs Analysis	Verify that a user can input their business challenges into the text box, submit the form, and successfully receive tailored HR module recommendations from the AI.	The system successfully processes the text, transitions from the "Analyzing..." state, and displays relevant HR module recommendations without errors.	The system successfully processed the user's prompt regarding scaling from 10 to 50 employees. It successfully navigated to the Recommendations screen and generated context-aware	Pass	 

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
				advice (mentioning EPF, SOCSO, etc.). It accurately selected 'Payroll & Statutory' and 'Time & Attendance' as 'AI Pick' modules, and successfully flagged the user for the MSME Digital Grant Madani 2025.		
TC-02	Specific Case (Negative): Empty Input Submission	Verify that the system handles an empty submission correctly when a user clicks "Analyzing" without typing any HR needs.	The system should detect that the input field is empty and prevent the form from submitting. It must block any blank requests from being sent to the AI API to prevent errors or wasted API calls. The application must remain stable and not crash.	Form submission was successfully blocked by frontend validation. The system prevented the empty API call from triggering, the 'Analyze & Recommend' button did not proceed to the next step, and the application remained perfectly stable.	Pass	

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
TC-03	NFR (Performance): AI API Latency	Measure the response time from the moment the user clicks the "Analyzing" button to the moment the UI renders the final recommendations from the API via Vercel.	The system must maintain an acceptable API latency for AI queries to ensure a responsive user experience. The average response time from the deployed backend endpoint to the AI API should remain under 4.0 seconds per request, with a 0% error rate.	The deployed AI endpoint performed within acceptable limits. Network panel tracking showed successful fetch requests returning with status 200. Observed response times were 3.70s and 2.74s, resulting in an average latency of approximately 3.22 seconds with 0 errors.	Pass	
TC-AI-01	AI Output Validation: AI Output Validation: Multi-language NLP (Manglish/BM) G	Send a prompt written in colloquial Manglish/BM (e.g., "Company very teruk now, staff always cabut") to ensure the AI understands the local context and does not hallucinate irrelevant modules.	The NLP engine should correctly interpret Malaysian slang and Manglish (e.g., 'teruk', 'cabut', 'pening kepala') as specific HR challenges related to rapid employee turnover and hiring costs. It must return relevant, professional HR strategies without	The system successfully parsed the Manglish input. The AI accurately identified 'cabut' as high turnover within the 90-day probation period. It provided actionable, professional solutions including structured	Pass	

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
			hallucinating or failing to understand the local context	onboarding, exit interviews, and salary benchmarking. Zero hallucinations were detected.		

2. Automated Testing Pipeline

2.1. Unit Testing (Core Logic & Functions)

- **Test Runner Used:** Jest
- **Targeted Components:** `parseAIResponse()`, `validateFormInput()`
- **File(s) Covered:**

<https://github.com/TeohXiXian-2025/umhackathon-1/blob/main/frontend/utils/helpers.js>

Unit Test ID	Function/Component Tested	Test Scenario	Expected Outcome	Actual Result	Status
UT-01	<code>parseAIResponse()</code>	Pass a mocked valid JSON string to the function to verify it correctly extracts the AI modules without crashing.	Returns the parsed modules array accurately.	Successfully extracted modules from valid JSON (Passed in 2ms).	Pass
UT-02	<code>validateFormInput()</code>	Test edge cases for user input by passing both an empty string and a valid string to the validation logic.	The function should correctly flag empty strings as invalid and valid strings as valid. t	Successfully blocked empty strings and allowed valid inputs.	Pass

Execution Proof:

```
(.venv) PS C:\Users\User\Documents\UM_HACKATHON\umhackathon\frontend> npm run test

> frontend@0.1.0 test
> jest

PASS utils\helpers.test.js
  AI Decision Support Helper Functions
    ✓ parseAIResponse should extract modules from valid JSON (2 ms)
    ✓ validateFormInput should block empty strings
    ✓ validateFormInput should allow valid input

Test Suites: 1 passed, 1 total
Tests: 3 passed, 3 total
Snapshots: 0 total
Time: 0.499 s
Ran all test suites.
```

2.2. Integration & API Testing (System Modules)

- **Test Tool Used:** Postman (API Client)
- **Targeted Integrations:** Next.js Frontend Client -> Next.js Backend API Route (/api/ai) -> AI Agent
- **File(s) Covered:**
<https://github.com/TeohXiXian-2025/umhackathon-1/blob/main/frontend/app/api/ai/route.js>

Test ID	Integration / Point Tested	Test Scenario	Expected Outcome	Actual Result	Status
IT-01	POST /api/ai -> API	Send a valid HTTP POST request with a JSON body (e.g., {"prompt": "Need to track attendance for 20 staff"}) to the API endpoint to verify successful communication with the external AI model.	Should return HTTP Status 200 OK. The response should be a structured JSON object containing the recommended HR modules from the AI.	Returned HTTP 200 OK. The JSON payload successfully returned 3 parsed module recommendations without timing out.	PASS

Execution Proof:

My Collection > Post data

POST <https://umhackathon-1-9n2r.vercel.app/api/ai> [Send](#)

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON [Schema](#) [Beautify](#)

```
1 {
2   "prompt": "My staff keep resigning after 3 months, how do I fix this?",
3   "systemInstruction": "You are a professional HR consultant for Malaysian SMEs. Recommend the best
4     PeopleGraph HR software modules to solve the user's problem. Be concise."
5 }
```

Body Cookies Headers 11 Test Results 1/1 [200 OK](#) • 6.05 s • 2.09 KB [Save Response](#)

Testing your API endpoints one by one? Run your entire collection and test end-to-end workflows effortlessly. [Run Collection](#)

Filter Results [PASSED](#) Successful POST request

My Collection > Post data

POST <https://umhackathon-1-9n2r.vercel.app/api/ai> [Send](#)

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON [Schema](#) [Beautify](#)

```
1 {
2   "text": "High turnover within the first 90 days usually stems from a \"disconnect\" between
3     expectations and reality. To stabilize your retention, I recommend these **PeopleGraph**
4     modules:\n\n### 1. Digital Onboarding Module\nFirst impressions are critical. This module ensures a
5     structured integration process rather than a \"sink or swim\" approach.\n\n**The Fix:** Automate
6     welcome workflows, digital document signing, and orientation checklists so new hires feel supported
7     and valued from Day 1.\n\n### 2. Performance Management (Probation Review)\nStaff often leave
8     because they aren't sure if they are doing a good job.\n\n**The Fix:** Use this module to set
9     clear **30-60-90 day KPIs**. Regular automated check-ins during the probation period allow you to
10    address concerns before the employee decides to resign.\n\n### 3. Employee Engagement (Pulse
11    Surveys)\nYou can't fix what you don't measure.\n\n**The Fix:** Deploy automated **\"New Hire
12    Surveys\"** at the end of Month 1 and Month 2. This captures honest feedback about their experience
```

Body Cookies Headers 11 Test Results 1/1 [200 OK](#) • 6.05 s • 2.09 KB [Save Response](#)

[JSON](#) [Preview](#) [Visualize](#)

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3. Defect Log & Fix Traceability (Root Cause Analysis)

Bug ID	Bug Description	Steps to Reproduce	Console/Error Log Snippet	Root Cause Analysis (Why did it break?)	Fix Commit Hash / PR Link
BUG-01	Backend API Route (/api/ai) crashed with a 500 Error.	1. Send POST to /api/ai. 2. Omit systemInstruction in JSON. 3. Observe server crash.	required one of field 'data' must have one initialized field	The backend expected systemInstruction from the client; receiving undefined caused the SDK to reject the payload.	c27dbf5
BUG-02	Submit button allowed multiple rapid clicks (Double Submit).	1. Enter valid HR prompt. 2. Rapidly double-click "Analyze". 3. Observe duplicate API calls.	[Network] 2 concurrent POST requests to /api/ai	Missing UI state management; the button was not disabled during the isLoading phase of the fetch request.	abb431b

4. Known Issues & Deferred Technical Debt

Bug ID	Description & Impact (Bug or Technical Debt)	Reason for Deferral	GitHub Issue Link
DEF-UI-01	UI Glitch: The Force-Directed Graph nodes sometimes drift out of the viewport on 4K monitors.	Minor CSS/SVG bounding box issue. Fixed for standard 1080p displays; deferred 4K optimization due to time constraints.	https://github.com/TeohXiXi/an-2025/umhackathon-1/issues/9#issue-4367730880
DEF-TD-01	Architectural Trade-off: The "Value Graph" Digital Twin currently fetches the entire employee dataset at once.	Acceptable for SMEs with <100 employees. For larger enterprises, pagination or "Lazy Loading" for D3 nodes would be required to prevent browser lag.	https://github.com/TeohXiXi/an-2025/umhackathon-1/issues/10#issue-4367736706
DEF-PG-01	Data Limitation: Statutory calculations currently do not support "Part-Time" or "Contractor" categories (Form CP159)	Focused on full-time Malaysian employees (EA Form) as they represent 85% of the target SME user base.	https://github.com/TeohXiXi/an-2025/umhackathon-1/issues/11#issue-4367740129

5. Live Demo / UAT Risks

- **Risk 1: Backend "Cold Start" Latency**

Description: Our backend services are hosted on a free-tier cloud environment (Railway/Render).

Operational Instruction: If the application has been inactive, the first API call (e.g., loading the Dashboard) may take 20-30 seconds to "wake up" the server. Please allow the initial loading spinner to complete without refreshing the page.

- **Risk 2: AI Simulation Processing Time (Cuti Peristiwa)**

Description: The "Cuti Peristiwa Shock Simulation" involves multi-layered ensemble forecasting and calls to the AI model.

Operational Instruction: After clicking "Run Simulation," the system may appear "frozen" for 8-12 seconds while it processes the unstructured data and performs calculations. Do not click the button multiple times; a success notification will appear once the analysis is complete.

- **Risk 3: Stateless Environment (Data Persistence)**

Description: To ensure rapid deployment on Vercel, this version uses a stateless architecture.

Operational Instruction: Any new employees added or settings changed during the demo are saved to a temporary session. Refreshing the browser tab will reset the system to its original "Demo Mode" state. We recommend following the demo flow without a hard refresh.

- **Risk 4: Visualization Display Density (D3 Value Graph)**

Description: The Force-Directed Value Graph is a high-compute Digital Twin visualization optimized for desktop processing.

Operational Instruction: This component was specifically designed for Google Chrome on Desktop (1920x1080 resolution). Viewing the graph on a mobile device or a small tablet may cause the nodes to overlap, making the "revenue-to-skill" lines difficult to distinguish.